

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 11: Areas Related to Circles

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Sector and segment of a circle

1. A sector with a central angle of 72° has an area of 40π square units. Find the radius of the circle.

CBSE 2025, 1 mark

SOLUTION

$$\frac{72}{360} \times \pi r^2 = 40\pi, \text{ so } \frac{r^2}{5} = 40$$

$$r^2 = 200$$

Answer: $r = 10\sqrt{2}$ units

2. A wire 20 cm long is bent into an arc of a circle of radius $\frac{60}{\pi}$ cm. What angle does the arc subtend at the centre? CBSE 2025, 1 mark

SOLUTION

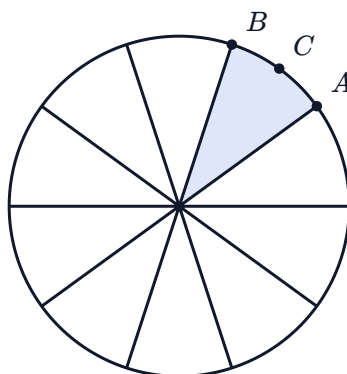
$$\text{Arc length} = \frac{\theta}{360} \times 2\pi r, \text{ so } 20 = \frac{\theta}{360} \times 2\pi \times \frac{60}{\pi} = \frac{\theta}{360} \times 120$$

$$\theta = \frac{20 \times 360}{120}$$

Answer: 60°

3. A circular ornament of diameter 35 mm is made from silver wire. Wire also forms 5 diameters, which cut the circle into 10 equal sectors. A and B are the ends of one sector on the circle and C lies on the arc between them. Take $\pi = \frac{22}{7}$. (i) Find the central angle of each sector. (ii) Find the length of the arc ACB . (iii) Find the area of each sector, OR the total length of wire used.

CBSE 2025, 4 marks



SOLUTION

- (i) Ten equal sectors share 360° . **Each central angle is 36° .**
- (ii) Radius $r = \frac{35}{2}$ mm, so the circumference is $2 \times \frac{22}{7} \times \frac{35}{2} = 110$ mm. Arc $ACB = \frac{36}{360} \times 110$.
Arc $ACB = 11$ mm.
- (iii) Area of the circle = $\frac{22}{7} \times \frac{35}{2} \times \frac{35}{2} = 962.5$ mm². Each sector is one-tenth of it. **Area of each sector = 96.25 mm².**
- OR: wire used = circumference + 5 diameters = $110 + 5 \times 35$. **Total wire = 285 mm.**

4. In a circle of radius 21 cm, an arc subtends 60° at the centre. Find (i) the length of the arc (ii) the area of the minor segment cut off by the chord of this arc. **CBSE 2024, 5 marks**

SOLUTION

(i) Length of arc = $\frac{60}{360} \times 2 \times \frac{22}{7} \times 21 = \frac{1}{6} \times 132 = 22$ cm

(ii) Area of sector = $\frac{60}{360} \times \frac{22}{7} \times 21 \times 21 = 231$ cm²

The triangle formed by the two radii and the chord has an angle of 60° between equal sides, so it is equilateral with side 21 cm. Its area = $\frac{\sqrt{3}}{4} \times 21^2 = \frac{441\sqrt{3}}{4}$ cm²

Area of minor segment = $\left(231 - \frac{441\sqrt{3}}{4} \right)$ cm² $\approx 231 - 190.95 = 40.05$ cm² (taking $\sqrt{3} = 1.732$)

Answer: (i) 22 cm (ii) $\left(231 - \frac{441\sqrt{3}}{4} \right)$ cm² ≈ 40.05 cm²

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